

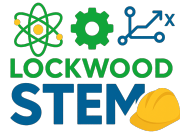
Introduction to Engineering Design

Aerospace Academy Version

Unit 4 Rubric

Introduction to Engineering Design

Course: Introduction to Engineering Design
Program: Academy of Advanced Aerospace Engineering
Brand: Lockwood STEM
Use: Student-facing and teacher grading rubrics



Unit 4

Measurement, Drawings & Manufacturing Documentation

Measurement tools, tolerances, technical drawings, manufacturing notes, BOMs, revision control, and fabrication package.

Grading Rubrics and Scoring Guide

Unit 4: Measurement, Drawings & Manufacturing Documentation

Assessment Purpose

Measurement tools, tolerances, technical drawings, manufacturing notes, BOMs, revision control, and fabrication package. This rubric is designed for consistent grading, feedback, and student self-checks.

Suggested 4-Point Scale

4 = Expert: exceeds expectations with clear evidence, precision, and independence. **3 = Proficient:** meets expectations with complete and accurate work. **2 = Emerging:** partially meets expectations; work is incomplete, vague, or inconsistent. **1 = Incomplete:** missing, unsafe, inaccurate, or shows limited evidence of understanding.

Unit 4 Score Summary

Category	Points Possible	Points Earned
Measurement technique	10	
Tolerances and fit	15	
Technical drawing views	15	
Dimensioning practice	15	
Manufacturing notes	10	
BOM and revision control	10	
Fabrication plan	10	
Documentation package quality	15	

Unit 4 Project Rubric



Criterion	Pts	4 – Expert	3 – Proficient	2 – Emerging	1 – Incomplete
Measurement Technique	10	Uses tools correctly, records units/precision, and verifies measurements with care.	Measurements are mostly correct with units.	Measurements are inconsistent or lack precision/units.	Measurements missing or unreliable.
Tolerances & Fit	15	Applies realistic tolerances and explains fit/function/manufacturing impact.	Tolerances are included and mostly reasonable.	Tolerances are vague, incomplete, or not connected to function.	No tolerances.
Technical Drawing Views	15	Views are complete, aligned, scaled, and clearly communicate geometry for manufacturing.	Views are mostly complete with minor issues.	Views are incomplete or contain geometry errors.	Drawing views missing.
Dimensioning Practice	15	Dimensions are complete, readable, non-redundant, and placed according to drafting expectations.	Dimensions mostly meet expectations.	Dimensions are incomplete, crowded, or duplicated.	Missing dimensions.
Manufacturing Notes	10	Notes clearly specify material, process, finish, safety/quality checks, and special instructions.	Notes include important manufacturing details.	Notes are incomplete or generic.	No manufacturing notes.
BOM & Revision Control	10	BOM is complete and revisions are clearly tracked with description/date/author.	BOM and revisions are mostly complete.	BOM or revision table is incomplete.	Missing BOM/revisions.
Fabrication Plan	10	Plan sequences steps logically and includes tools, materials, checks, and risks.	Plan is workable with minor gaps.	Plan is vague or missing steps.	No fabrication plan.
Documentation Package Quality	15	Final package is organized, professional, printable, and ready for review/manufacturing discussion.	Package is complete with minor issues.	Package is incomplete or not ready for use.	Package missing.

Introduction to Engineering Design
Aerospace Academy Version

Rubrics

Student Self-Check Before Submission

Requirement	Complete?
☐ Measurement technique evidence is complete and ready to be graded.	
☐ Tolerances and fit evidence is complete and ready to be graded.	
☐ Technical drawing views evidence is complete and ready to be graded.	
☐ Dimensioning practice evidence is complete and ready to be graded.	
☐ Manufacturing notes evidence is complete and ready to be graded.	
☐ BOM and revision control evidence is complete and ready to be graded.	
☐ Fabrication plan evidence is complete and ready to be graded.	
☐ Documentation package quality evidence is complete and ready to be graded.	
☐ All pages/files are named, organized, and submitted correctly.	
☐ Reflection or revision notes are complete.	

Teacher
Use the
evidence
those b